Lesson Plan — Session 1 · Customary Conversions (feet → inches)

Mr. Neft · Grade 6 Mathematics · Monday, October 5, 2026 · 90-minute block

Reveal Math: Ratio Reasoning: Convert Measurements within the Same System · Source: 3.6.pptx slides 1–21 and 39 · Practice/3.6 Session 1.docx · HW/3.6 Session 1 HW.docx

1. DO NOW

Do Now — Warm-Up (Slide 1) · 4 minutes	**Talk stem (TWR)**
Students work silently, then mark one answer they are unsure of. 1. 12 × 4 = _____ 2. 1 foot = _____ inches 3. 1,000 × 3 = _____ 4. 60 ÷ 12 = _____	*"Item ___ will help me today because 1 foot = ___ inches."* **Key:** 48 · 12 · 3,000 · 5

2. STANDARDS

Grade 5 Prerequisite	Grade 6 Target Standard	Grade 7 Extension
5.MD.A.1 — Convert among different-sized standard measurement units within a given measurement system (e.g., 5 cm to 0.05 m) and use these conversions in multi-step, real-world problems.	**Maryland 6.AT.3** (deck, slides 3 and 24) · **CCSS 6.RP.A.3d** (HW and Answer Key) — Use ratio reasoning to convert measurement units; manipulate and transform units appropriately when multiplying or dividing quantities. Deck target (slide 39): *I can use ratio reasoning to convert between units of the same measurement system.*	**7.RP.A.1** — Compute unit rates associated with ratios of fractions, including ratios of lengths, areas, and other quantities measured in like or different units.

3. OBJECTIVES

Objective	Statement	Self-check
Content	I can **convert** feet to inches by scaling the **unit ratio** 12 inches : 1 foot in a ratio table, and use the result to compare two heights.	☐ Before | ☐ After
Language (WIDA)	I will explain why the number changes but the height does not, using the frame "The units changed from ___ to ___, but the amount stayed the same" to justify my conversion to a partner. Sentence Frame: "I know this because..."	☐ Before | ☐ After

4. LESSON PROCEDURES

Phase (Time)	Teacher Moves	Student Moves	SPED Supports	Level 1 / WIDA Scaffolds	Formative Check
Warm-Up (8 min)	• Display the Warm-Up (Slide 1). Four items solo; students star one item they are unsure of. [T3] • Circulate and note who writes 1 foot = 10 inches or who multiplies on item 4 (60 ÷ 12). • Show More or Less Than 10 (Slide 12): each student picks one sum and defends an estimate. All three are less than 10 (9.53, 9.243, 9.803). [T4★] • Go over the answers 48 · 12 · 3,000 · 5 and ask which item helps change feet to inches.	• Solve items 1–4 alone; star one unsure answer and revise it in a second color. • Defend an estimate to a partner: *"I think 4.509 + 5.294 is ___ than 10 because 4.5 + 5.3 is about ___."* • Name the link: *"Item ___ helps because 1 foot = ___ inches."*	• R.M.G. — visual schedule: today's 8-phase agenda posted and pointed to before starting. • J.L. — clear expectations: "4 problems · 4 minutes" timer visible. • A.M.M. — wait time; teacher checks item 2 privately before whole-class sharing.	• L1–2: Point to 12 on a ruler to show 1 foot = 12 inches; answer items with numbers only. • L3–4: Answer and say "I multiplied/divided because ___." • L5: Explain which Warm-Up item connects to changing feet into inches.	• [T4] Scan items 2 and 4 on every paper; tally who has 1 ft = 12 in and 60 ÷ 12 = 5. • [T3] "How many inches are in 1 foot? How do you know?"
Launch (7 min)	• Show the Learning Target (Slide 3); students read the content and language objectives chorally. [T3] • Run Vocabulary 1–4 (Slides 4–7): convert measurements within the same system, conversion factor, unit ratio, convert. For each: "Where does this word appear in the example?" [T4] • Have students write today's conversion factor, 1 ft = 12 in, at the top of the note sheet.	• Restate the target to a partner: "Today I will convert ___ to ___ and explain why the ___ stays the same." Sentence Frame: "I know this because..." • Point to each word in its example: *"The conversion factor is 1 foot = ___ inches, so the unit ratio of inches to feet is ___ : 1."*	• J.C. — pre-teaching: convert / conversion factor / unit ratio cards previewed before class. • M.E. — word bank (convert, conversion factor, unit ratio, customary) on desk. • S.D.L.P. — verbal response: says each definition to the teacher instead of writing it.	• L1–2: Match word cards to pictures (ruler, 12 : 1); repeat each term aloud. • L3–4: Use the conversion-factor frame for each term. • L5: Explain how a conversion factor and a unit ratio are different. Sentence Frame: "I know this because..."	• [T4] Listen to partner restatements for "the amount (height) stays the same." • [T3] "What is the unit ratio of inches to feet?" (12 : 1)
Be Curious (10 min)	• Show Be Curious (Slide 11): 30 seconds of quiet looking at the bag on the scale, then collect "What is the question?" ideas. [T3] • Launch Riding the Attractions (Slide 13): the sign reads 44 in.; the child is 4 feet tall. Students predict with the slide frame. [T4★] • A/B: A states a prediction; B asks, "Which unit will you use?" Switch roles. • Do not confirm the answer. Record 2–3 predictions (for example, "4 is less than 44, so no") to revisit in Guided Practice.	• Write one question about the photo: *"I wonder how many ___ the bag weighs if the scale shows ___."* • Predict (Slide 13 frame): *"I need to compare ___ and ___ in the same unit. I can start with ___."* • Ask the partner for evidence: *"Why did you change ___ into ___?"*	• R.B. — simplified language card: "Child: 4 feet. Sign: 44 inches. Same unit?" • J.L. — movement: students stand at the classroom height chart to show 4 feet. • A.M.M. — safe response: predictions go on unsigned sticky notes.	• L1–2: Point to "44 in." and "4 feet"; sort them as same unit or different units. • L3–4: Use the Slide 13 prediction frame. • L5: Predict and justify why comparing 4 to 44 is not a fair comparison.	• [T4] Sort the sticky notes and count predictions that compare 4 directly to 44. • [T3] "What units are in this problem?" (feet and inches)

Phase (Time)	Teacher Moves	Student Moves	SPED Supports	Level 1 / WIDA Scaffolds	Formative Check
Guided Practice (15 min)	• Reveal sequence (Slides 19–20): gather attempts before each reveal. Units are different; 1 foot = 12 inches. **[T3]** • Collaborate and Connect (Slide 21): the ratio of inches to feet is 12 : 1, a unit ratio. "How can you use the unit ratio to convert 4 feet to inches?" A explains, B asks for evidence. **[T4★]** • Feet → Inches (Slide 14): build the ratio table 1 : 12, 2 : 24, 4 : 48; scale both rows by 4. Name the tools with Vocabulary 6–7 (Slides 9–10). • Solution (Slide 15): 48 inches > 44 inches, so the child can ride. Return to the recorded predictions. Copy the 3 note lines (Slide 2), cover one, and explain it.	• Complete the note-sheet ratio table: Feet 1, 2, 3, 4 → Inches 12, ___, ___, ___. • Explain the scaling (Slide 14 frame): *"I multiplied 1 foot and 12 inches by ___. The height stayed ___."* • Prove the answer: *"4 feet = ___ inches, and ___ inches is more than 44 inches, so the child ___ ride."*	• S.D.L.P. — manipulatives: four 12-inch rulers laid end to end to model 4 feet = 48 inches. • R.B. — graphic organizer: pre-printed two-row ratio table with unit labels. • J.C. — extended time for note copying; Slide 2 lines provided as a printed strip.	• L1–2: Fill the ratio table with number cards; say "4 feet equals 48 inches." • L3–4: Use the Slide 14 frame. • L5: Explain why multiplying both quantities by 4 keeps an equivalent ratio. Sentence Frame: "I know this because..."	• **[T4]** Check every table for 24 and 48 in the inches row, aligned under 2 and 4. • **[T3]** "Why do we multiply both 1 and 12 by 4, not just one of them?"
Collaborative Practice (15 min)	• Practice: Customary Conversions (Slide 16): 5 quarts to fluid ounces (1 qt = 32 fl oz), then write ounces : pounds and pounds : ounces. **[T4★]** • Assign Practice sheet items 2–4 (Field Day Ribbon, Party Punch, Gym Rope) A/B: A computes; B checks a second way. • Discuss the task (Slide 17): "Why does the number change while the actual measurement stays the same?" [T4] Sentence Frame: "I know this because..."	• Slide 16: 5 × 32 = 160 fl oz; *"The ratio ___ compares ounces to pounds. Reversing the order gives ___."* (16 : 1 → 1 : 16) • Items 2–4 (15 ft, 180 in · 12 qt, 48 cups · 24 ft, 288 in). Partner check: *"1 yard is ___ inches, so 5 yards is 5 × ___ = ___ inches."* • Slide 17: *"The number got bigger because inches are ___ than feet, but the length stayed ___."*	• J.L. — movement: partners switch seats between items 2 and 3. • R.B. — graphic organizer: arrow chain yd → ft → in labeled ×3 and ×12. • S.D.L.P. — calculator for the two-step products (5 × 3 × 12; 8 × 3 × 12).	• L1–2: Partner reads the problem aloud; student fills the table cells and labels the units. • L3–4: Use the partner-check frame. • L5: Explain why two steps (×3, then ×12) equal one step (×36). Sentence Frame: "I know this because..."	• **[T4]** Listen for B partners checking with 36 inches per yard. • **[T3]** "How many cups are in 1 gallon?" (4 × 4 = 16)
Independent Practice (12 min)	• Launch the three stations in Section 7. Post the rotation chart before releasing students. • Lead the Reteach table (Practice items 1, 10, 11) with rulers and the ratio table. **[T4]** • At minute 6, check the Practice station's item 7 (24 quarts → gallons), the first item that requires dividing.	• Work the assigned station set; label units on every step. • Before each item: *"___ are smaller than ___, so the number should get ___, so I ___."*	• M.E. — extended time with a core set (items 1, 10, 11); sentence starters on the table. • A.M.M. — 1:1 check-in after the first item; reassurance before moving on. • R.M.G. — visual station schedule and predictable rotation signal.	• L1–2: Items marked with a big-to-small arrow picture; answer with number and unit. • L3–4: Use the direction frame before each item. • L5: Item 13 — explain Sam's error in writing.	• **[T4]** Check the two dividing items (7 and 11) for every student at the Practice station. • **[T3]** "Is a pound bigger or smaller than an ounce? So will 80 get bigger or smaller?"

Phase (Time)	Teacher Moves	Student Moves	SPED Supports	Level 1 / WIDA Scaffolds	Formative Check
Closing (8 min)	• 60-Second Check (Slide 18): a child is 5 feet tall; convert to inches, explain, and self-monitor (READY / ALMOST / NOT YET). **[T4★]** • Learning Targets (Slide 39): students rate themselves on the first target. • Assign HW/3.6 Session 1 HW; preview Session 2 (liters and milliliters).	• Convert 5 ft = 60 in with a ratio table or equation. • Explain: *"The number changed from ___ to ___ because each foot holds ___ inches, but the height stayed the same."* • Circle READY, ALMOST, or NOT YET and ask a partner, "How do you know?"	• J.C. — verbal response for the explain part. • J.L. — reinforcement: specific praise for a revised answer. • R.M.G. — predictability: Session 2 agenda shown before dismissal.	• L1–2: Show the answer on a whiteboard; point to READY, ALMOST, or NOT YET. • L3–4: Use the explain frame. • L5: Explain to the class, pointing to the ratio table as evidence.	• **[T4]** Sort 60-Second Checks into READY / ALMOST / NOT YET and compare each self-rating to the work. • **[T3]** "What do you multiply by to change feet to inches?" (12)
Exit Ticket (15 min)	• Assign exit-ticket tiers (Section 9) using the 60-Second Check piles. • Silent and independent; read aloud on request. • Collect and sort to form Session 2 Reteach and Enrichment groups. **[T4]**	• Complete the assigned tier; show the conversion and label units. • Tier 1 frame: *"I ___ by 12 because inches are ___ than feet. 6 feet = ___ inches."*	• M.E. — extended time and word bank. • S.D.L.P. — calculator. • J.C. — text-to-speech or teacher read-aloud. • A.M.M. — 1:1 check-in and reassurance before submitting.	• L1–2: Tier 1 read aloud with a ruler picture. • L3–4: Tier 2 with the explain frame. • L5: Tier 3.	• **[T4]** Score each ticket against the Section 9 key. • **[T3]** On collection: "Did you multiply or divide? Why?" • **Target: 80%+ convert 6 ft = 72 in and state they multiplied because inches are smaller than feet.**

Total: 90 minutes. TEACH codes: **[T3]** ask–respond–affirm · **[T4]** justify · **[T4★]** justify + peer discourse or written reasoning.

5. VOCABULARY

Tier	Term	Student-Friendly Definition	Morphology (root + affix)	EN-ES Cognate ✓/✗	Cross-Disciplinary Use
3	customary units	The U.S. units: inch, foot, yard, mile, ounce, pound, cup, quart, gallon.	custom + -ary (relating to)	✗ (unidades usuales)	Cooking and recipes (cups, ounces); U.S. road signs (miles)
2	convert	Write the same amount using a different unit.	con- (together) + vert (turn)	✓ convertir	Science: convert grams to kilograms; money: convert dollars to cents
3	conversion factor	The pair of equal amounts you multiply or divide by, such as 1 ft = 12 in.	convert + -sion; fact (make) + -or	✓ factor de conversión	Science lab measurement; currency exchange rates
3	unit ratio	A ratio that compares an amount to exactly 1 unit, like 12 inches to 1 foot.	unit (Latin unus, one) + ratio (reckoning)	✓ razón unitaria (unitaria ✓)	Unit price in shopping; speed in miles per hour
3	ratio table	A table of equal ratios that scales a conversion up or down one column at a time.	ratio (reckoning) + table	✓ tabla de razones	Recipes scaled for more people; science data tables
2	measurement system	A family of units that go together — customary or metric.	measure + -ment; syn- (together) + stem (stand)	✓ sistema de medición	Science (metric); geography (map scales)

6. MISCONCEPTIONS + INTERVENTIONS

#	Student Belief	Cognitive Root	Teacher Corrective Move	Intervention Route
1	Converting inches to feet, multiply by 12 (Sam: 60 × 12 = 720 feet).	Memorized "use 12" without reasoning about unit size; no reasonableness check.	Ask: "Are feet bigger or smaller than inches? Should there be more or fewer of them?" Model 60 in on rulers → 5 groups of 12.	Practice item 13 (Sam's Work); Reteach station item 11 (80 oz → lb).
2	40 inches is more than 4 feet because 40 > 4.	Compares the numbers and ignores the units.	Return to Riding the Attractions (Slides 19, 13): change to the same unit first, then compare 48 and 40.	Slide 19 revisit; HW item 8 (Elijah's ribbon).
3	Scale only one quantity in the ratio table (1 : 12 → 4 : 12) or add instead of multiply.	Treats the table as two separate lists, not paired equivalent ratios.	Point to Slide 14: circle a column and say "both rows times 4." Check with rulers: 4 rulers = 48 in.	Slide 14 table; Reteach station item 1b (Roller Coaster table).
4	ounces : pounds and pounds : ounces are the same ratio.	Order of quantities in a ratio is not yet attached to meaning.	Label each number with its unit; read aloud: "16 ounces for every 1 pound" vs "1 pound for every 16 ounces."	Slide 16 item 2; Collaborative Practice.
5	Stops after one step in a two-step conversion (5 yd = 15, done).	Loses track of the target unit across two conversion factors.	Write the unit chain yd → ft → in above the work and check off each arrow.	Practice items 2, 4, 12; Practice-station partner check.

7. SMALL-GROUP STATIONS (RUN DURING INDEPENDENT PRACTICE)

Station	Who	Problems	Guiding Question / Protocol	Expected Artifact	TWR Frame
RETEACH (teacher-led)	M.E., R.B., J.C., R.M.G., J.L., S.D.L.P., A.M.M. (full accommodations) + students who placed NOT YET on the Warm-Up	Practice sheet item 1 (Roller Coaster Height Limit: 48 in rule, Caleb 4 ft) · item 10 (Bookshelf 6 ft → 72 in) · item 11 (Bag of Flour 80 oz → 5 lb)	"Is the new unit bigger or smaller? So will the number grow or shrink?"	Labeled ratio table for item 1 (4 ft = 48 in, Caleb can ride) plus two direction-labeled conversions.	*"Feet are ___ than inches, so I ___ by 12. 4 feet = ___ inches."*
PRACTICE (partner)	Partners (A/B), mixed READY / ALMOST	Practice sheet items 5–9 and 12 (4.5 lb → 72 oz · 2.5 mi → 13,200 ft · 24 qt → 6 gal · 6.5 lb → 104 oz · 9 gal → 36 qt → 72 pint cups · 7 yd → 21 ft → 252 in)	A solves; B checks the direction (multiply or divide) before checking the number. Switch every item.	Each item answered with units, plus B's initials beside every check.	*"You should ___ because ___ are bigger than ___, so there will be ___ of them."*
ENRICHMENT (independent)	Independent, READY on the Warm-Up and 60-Second Check	Practice item 13 (Sam's Work) · item 14 (Fencing a Garden: 24 ft → 8 yd, $72; by the foot 24 × $3.25 = $78)	Why is buying by the foot more expensive, even though the fence is the same length?	A written paragraph comparing the two prices with labeled conversions.	*"Buying by the yard costs ___ because 24 feet = ___ yards. At $3.25 per foot, 1 yard costs ___, which is ___ than $9."*

8. STUDENT NOTE SHEET

Lesson 3-6 · Converting Within One System · Session 1 Notes Name _____ Date _____ Period ____

A · Warm-Up Link

1 foot = _____ inches

60 ÷ 12 = _____ This helps me change _____ into _____.

B · Riding the Attractions (Slides 13–15)

Sign: YOU MUST BE THIS TALL TO RIDE — 44 in. Child: 4 feet

My prediction: "I need to compare ___ and ___ in the same unit. I can start with ___."

Feet	1	2	3	4
Inches	12	___	___	___

4 feet = _____ inches. _____ inches > 44 inches, so the child _____ ride.

C · My 3 Lines (Slide 2)

01 Convert means write the _____ amount in another _____.

02 Start with a known _____ _____.

03 Use _____ _____ to find the new number.

D · Which Way? (Slide 16)

Bigger unit → smaller unit: _____ (5 quarts = _____ fluid ounces, 1 qt = 32 fl oz)

Smaller unit → bigger unit: _____ (80 ounces = _____ pounds, 1 lb = 16 oz)

E · Explain It (TWR)

"A foot is _____ than an inch, so for the same height the number of inches is always _____ than the number of feet, because _____."

Goal tracker: I can convert customary units using a unit ratio and explain why the amount stays the same. Sentence Frame: "I know this because..."	☐ 1 — Need help	☐ 2 — Need a hint	☐ 3 — Got it	☐ 4 — Can teach it

9. EXIT TICKET

Tier	Task	Frame	Answer Key
Tier 1 · Identify + frame	Convert 6 feet to inches. (1 foot = 12 inches) Circle: multiply / divide.	*"I___ by 12 because inches are ___ than feet. 6 feet = ___ inches."*	Multiply · smaller · 72 inches.

Tier	Task	Frame	Answer Key
Tier 2 · Identify + explain (TWR 2–3 sentences)	Adapted from Riding the Attractions (Slide 13): A ride requires riders to be at least 4 feet tall. Ana is 50 inches tall. Can she ride? Show the conversion, then explain in 2–3 sentences.	*"First I changed ___ to ___ because ___. Then I compared ___ and ___. So Ana ___ ride."*	4 × 12 = 48 inches; 50 > 48, so yes, Ana can ride. The explanation must change to one unit before comparing.
Tier 3 · Multi-step + contrastive + transfer	Banner Fabric (Practice item 23): A banner needs 7 feet of fabric. Fabric is sold by the yard (3 ft = 1 yd). (a) Write 7 feet in yards. (b) How many whole yards must you buy? (c) Jordan says buy 2 yards because 2 1/3 rounds to 2. Explain who is right, using feet as evidence. (d) Transfer: How would your method change for 7 pints of juice sold only by the quart (2 pt = 1 qt)?	*"Jordan is ___ because 2 yards is only ___ feet, which is ___ than 7 feet."*	(a) 7 ÷ 3 = 2 1/3 yd. (b) 3 yards. (c) Jordan is wrong: 2 yd = 6 ft < 7 ft. (d) Same move, dividing by 2: 7 ÷ 2 = 3 1/2 qt, so buy 4 quarts.

10. GRADE 6 IEP ACCOMMODATIONS MATRIX

Student	Full Accommodation List	Applied This Session (phase: support)	Small Group
M.E.	Sentence starters, word banks, chunking, extended time, manipulatives	• Launch: word bank (convert, conversion factor, unit ratio, customary) on desk. • Independent Practice: extended time with a core set (items 1, 10, 11); sentence starters on the table. • Exit Ticket: extended time and word bank.	Reteach station (all 7 students)
R.B.	Graphic organizers, simplified language, small group, breaks	• Be Curious: simplified language card: "Child: 4 feet. Sign: 44 inches. Same unit?" • Guided Practice: graphic organizer: pre-printed two-row ratio table with unit labels. • Collaborative Practice: graphic organizer: arrow chain yd → ft → in labeled ×3 and ×12.	Reteach station (all 7 students)
J.C.	Text-to-speech, verbal responses, extended time, pre-teaching	• Launch: pre-teaching: convert / conversion factor / unit ratio cards previewed before class. • Guided Practice: extended time for note copying; Slide 2 lines provided as a printed strip. • Closing: verbal response for the explain part. • Exit Ticket: text-to-speech or teacher read-aloud.	Reteach station (all 7 students)
R.M.G.	Visual schedules, sensory mods, motor mods, predictability	• Warm-Up: visual schedule: today's 8-phase agenda posted and pointed to before starting. • Independent Practice: visual station schedule and predictable rotation signal. • Closing: predictability: Session 2 agenda shown before dismissal.	Reteach station (all 7 students)
J.L.	Movement, clear expectations, reinforcement, chunking, visuals	• Warm-Up: clear expectations: "4 problems · 4 minutes" timer visible. • Be Curious: movement: students stand at the classroom height chart to show 4 feet. • Collaborative Practice: movement: partners switch seats between items 2 and 3. • Closing: reinforcement: specific praise for a revised answer.	Reteach station (all 7 students)
S.D.L.P.	Multi-sensory, manipulatives, verbal responses, calculator	• Launch: verbal response: says each definition to the teacher instead of writing it. • Guided Practice: manipulatives: four 12-inch rulers laid end to end to model 4 feet = 48 inches. • Collaborative Practice: calculator for the two-step products (5 × 3 × 12; 8 × 3 × 12). • Exit Ticket: calculator.	Reteach station (all 7 students)
A.M.M.	Wait time, safe responses, 1:1 check-ins, reassurance	• Warm-Up: wait time; teacher checks item 2 privately before whole-class sharing. • Be Curious: safe response: predictions go on unsigned sticky notes. • Independent Practice: 1:1 check-in after the first item; reassurance before moving on. • Exit Ticket: 1:1 check-in and reassurance before submitting.	Reteach station (all 7 students)